HX-520 Hygienic Installation and Operating Manual

Controlled technical manual

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1. Scope and safety

This manual applies to the HX-520 Hygienic CIP Pump in standard Helios configurations.

Installation, commissioning and maintenance must be performed by qualified personnel. Isolate electrical and process energy before service.

2. Operating envelope

Parameter	Limit / guidance
Flow range	10-100 m³/h
Head	up to 70 m
Maximum system pressure	12 bar
Pump-body continuous temperature	135°C continuous; 145°C for short sterilisation cycles up to 30 minutes
Suspended solids	Not intended for abrasive solids; particles should remain below 1 mm

3. Seal configuration

Standard EPDM hygienic seal is not recommended above 125°C continuous.

Standard configuration: FDA-compliant EPDM hygienic seal. Standard continuous-temperature guidance: 125°C continuous.

Available alternatives: FKM hygienic seal up to 140°C continuous.

Seal life is influenced by fluid chemistry, shaft speed, pressure, temperature cycling and dry-running events.

4. Installation

• Install on a rigid, level foundation.

• Align coupling after piping loads are connected.

• Avoid unsupported pipework loads at suction and discharge nozzles.

• Provide adequate suction conditions and avoid operation outside the recommended flow region.

• Confirm direction of rotation before introducing process fluid.

5. Commissioning checklist

Check	Acceptance criterion
Process isolation removed	Confirmed safe to open
Pump primed	Casing full; no dry running
Seal flush / support	Configured if required
Coupling alignment	Within installation tolerance
Temperature and pressure	Within documented limits

6. Maintenance

Hygienic seal inspection every 1,500 operating hours or per validated plant maintenance plan

Record operating temperature, pressure, vibration observations and seal leakage during planned inspections.

7. Troubleshooting

Symptom	Likely causes	First checks
Low flow	Incorrect rotation; restriction; air ingress	Check rotation, valves, suction line
High vibration	Misalignment; cavitation; pipe strain	Check alignment, NPSH conditions, supports
Seal leakage	Seal wear; incompatible material; dry running	Review service conditions and seal selection
Overheating	Operation outside envelope; poor cooling	Check duty point and fluid temperature